

Structure-Resolved Hybrid Modelling of CO₂ Adsorption in DEM-Generated Packed Beds

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Abstract

Packed-bed adsorption depends on sorbent chemistry and on the void pathways through which gas reaches the particles. Beds with nearly equal bulk porosity can therefore exhibit different transient utilization. This study develops an isothermal framework containing no CFD. The discrete-element method generates twenty mechanically realizable beds of monodisperse commercial zeolite 13X beads. A geometry-defined pore network is extracted from each packing, and a conservative transient solver couples gas-phase CO₂ diffusion through pore throats to one linear-driving-force adsorption equation per bead. All cases use the same geometry scale, material parameters and finite-area inlet condition; only the stochastic packing changes. At 5000 s, uptake varies from 0.206 to 0.315 mol kg⁻¹ although bulk porosity occupies only a narrow range. Porosity-only leave-one-bed-out models have negative predictive R^2 . In contrast, a physically defined steady conductance from the inlet to a plane five particle diameters inside the bed correlates strongly with uptake ($\rho = 0.968$) and underutilized-particle fraction ($\rho = -0.959$). The number of distinct second-layer pores is most closely associated with the relative half-uptake time ($\rho = -0.814$). The current paper therefore focuses on interpretable inlet accessibility and internal transport rather than graph neural networks. The next reduction step embeds selected physical descriptors in a homogeneous one-dimensional diffusion-adsorption model. The intended contribution is quantitative evidence that bulk porosity alone cannot describe transient sorbent utilization in narrow stochastic beds, together with a reproducible DEM-pore-network route to a more informative closure.

Keywords: carbon dioxide adsorption; zeolite 13X; packed bed; discrete-element method; pore network; inlet accessibility; hybrid modelling; linear driving force

1 Motivation, research gap and hypothesis

Fixed beds of porous adsorbent particles are widely considered for CO₂ separation. At the material scale, equilibrium capacity and intracrystalline or macropore transport determine how much and how rapidly CO₂ can be stored. At the bed scale, the arrangement of the beads determines the connected void space through which CO₂ reaches the sorbent. Standard column models average the second contribution into a bed porosity, an axial-dispersion coefficient and an effective kinetic constant. This is efficient but cannot distinguish two beds with equal global descriptors and different local bottlenecks.

Particle-resolved flow simulations could resolve this heterogeneity, but they would dominate the cost of producing hundreds of independent packings. The present proposal instead uses a geometry-defined transport network. Pore-network models replace the continuous void phase by pore bodies connected by throats and can enforce local conservation at much lower cost than a resolved continuum calculation [1]. The network is not an arbitrary graph: its nodes, edges, lengths and cross-sections are extracted from a mechanically realizable DEM packing.

The present reduction uses transparent physical descriptors rather than a learned graph representation. It identifies structural information that is lost when a packing is reduced to porosity or mean particle size and expresses that information through a small number of closure variables for a cheaper model. The central hypothesis is:

For DEM beds with matched particle-size distribution and global porosity, diffusion-limited CO₂ uptake still varies because of differences in inlet accessibility, pore connectivity and throat bottlenecks. A physically defined inlet-to-interior conductance and a small number of spatial descriptors capture this variation more reliably than global porosity alone.

This hypothesis is falsifiable. The proposed descriptors are unnecessary if porosity and a conventional tortuosity estimate predict withheld beds with comparable accuracy. Models are therefore compared by leaving out one complete packing at a time, not by reporting training accuracy.

2 Scope and physical interpretation

The recommended first paper addresses an initially CO_2 -free, nitrogen-filled bed exposed to a prescribed CO_2 concentration at its lower boundary. CO_2 penetrates the initially stagnant gas by diffusion and is adsorbed by the beads. The upper boundary is closed or assigned a specified concentration depending on the verification test. There is no imposed gas velocity, Navier–Stokes solver or CFD component. Consequently, the principal outputs are transient uptake, penetration depth and sorbent utilization, not a flowing-column breakthrough curve.

This restriction is important. A diffusion-only network is a defensible first test of how packing geometry influences transport and adsorption. It must not be described as a complete pressure- or vacuum-swing adsorption column model. A later extension can add a pressure-resistance network and throat advection, still without CFD, but that extension should follow validation of the present diffusion–adsorption solver.

Figure 1 illustrates the physical problem and the two coupled graphs. DEM particles define the solid skeleton. The dual pore network carries gas-phase CO_2 , while every adsorbing particle retains its own solid loading.

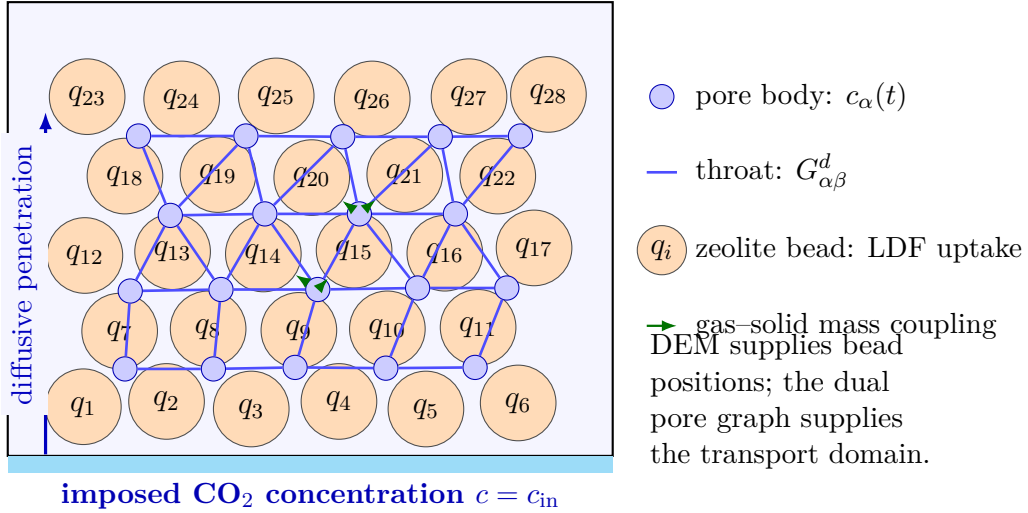


Figure 1: Schematic of the proposed diffusion–adsorption problem. Gas-phase CO_2 concentrations c_α are solved on pore bodies connected by geometry-derived throat conductances. Each zeolite bead has a separate loading q_i governed by an LDF equation. Gas removed by adsorption appears with equal and opposite sign in the neighboring pore balances.

3 Choice of sorbent and model parameters

3.1 Why zeolite 13X is the recommended first material

Commercial zeolite 13X is selected for four reasons. First, it is a benchmark physical adsorbent for post-combustion CO₂ separation and exhibits strong CO₂/N₂ selectivity [2, 3]. Second, commercial 13X is available as nearly spherical millimetre-scale beads, making the relationship between the laboratory material and a spherical DEM representation unusually direct. Hu et al. studied commercial beads with radii of approximately 0.8, 0.98 and 1.95 mm and found that CO₂ uptake under representative conditions was governed primarily by macropore diffusion [3]. This supports the use of a bead-scale LDF kinetic law. Third, equilibrium and transient data are available for checking the adsorption submodel. Dantas et al. demonstrated that a Toth equilibrium model combined with LDF kinetics can reproduce CO₂ and CO₂/N₂ fixed-bed adsorption on 13X [2]. Fourth, using a single established sorbent isolates the proposed contribution: the influence of bed geometry, rather than a comparison of sorbent chemistries.

Activated carbon is not chosen for the pilot because its surface chemistry and pore-size distributions vary strongly between feedstocks and activation routes. Amine-functionalized solids and reactive pellets are also postponed because humidity sensitivity, reaction kinetics and thermal effects would make it difficult to identify whether an observed trend comes from bed structure or material chemistry.

3.2 Reference properties and their status

Table 1 distinguishes physical inputs from numerical DEM choices. This distinction is essential. The density, bead size and adsorption data should correspond to a selected commercial 13X dataset. By contrast, the DEM Young’s modulus may be reduced to increase the stable timestep, provided the maximum overlap and final packing statistics are shown to be insensitive to further stiffness increases.

The baseline bed should contain approximately 1000–3000 particles. Smaller 200–500 particle beds are useful for debugging, mesh and exact-conservation tests, but can exaggerate boundary effects. Cylinder-to-particle diameter ratios of 10, 15 and 20 are proposed so that wall-induced packing structure can be quantified. A bed-height-to-diameter ratio of approximately 2 is sufficient for the first diffusion-uptake problem

Table 1: Proposed baseline properties for the first study. Adsorption parameters are fitted as one internally consistent set to a chosen 13X isotherm/uptake dataset; values from different products are not mixed.

Quantity	Baseline/range	Choice and justification
Sorbent	commercial zeolite 13X	benchmark CO ₂ adsorbent; strong CO ₂ /N ₂ selectivity; documented bead-scale kinetics
Bead diameter	1.6 mm baseline; 1.4–2.0 mm sensitivity	1.6 mm corresponds to the commercial $R \approx 0.8$ mm bead studied by Hu et al.; a narrow range tests polydispersity without changing material
Particle shape	sphere	consistent with approximately spherical commercial beads and provides an unambiguous radical-Voronoi pore construction
Apparent bead density	approximately $1.6 \times 10^3 \text{ kg,m}^{-3}$, verified for the selected batch	consistent with the reported 3.5 mg mass and 0.8 mm radius as an approximate inference; final simulations should use directly reported or measured batch data
Solid density for adsorption conversion	1950 kg,m^{-3} where the Dantas dataset is used	reported for the 13X sorbent in the selected fixed-bed study; it must not be silently interchanged with apparent bulk-bed density
Poisson ratio	0.25	representative numerical value for a brittle porous bead; low influence on the final gravity-packed geometry
DEM Young’s modulus	10^8 Pa baseline; 10^7 – 10^9 Pa sensitivity	intentionally softened numerical stiffness; accepted only when normalized overlaps and packing descriptors are stiffness independent
Particle friction	0.4 baseline; 0.2–0.6 sensitivity	spans plausible smooth-to-rough granular packing behaviour and generates controlled structural variability
Restitution coefficient	0.3	promotes rapid settling; it affects the preparation transient rather than the static adsorption equations
Rolling-friction coefficient	0.02 baseline; 0–0.05 sensitivity	represents resistance caused by bead roughness and nonsphericity not resolved by perfect spheres
Temperature	311 K baseline	matches the 38°C kinetic condition reported by Hu et al.; the first network model is isothermal
Feed CO ₂ partial pressure	0.1 bar	representative dilute post-combustion condition used in the single-bead kinetic experiments
Intrabead porosity and tortuosity	product-specific; e.g. $\varepsilon_p = 0.269$, $\tau_p = 2.61$ for $R = 0.98 \text{ mm}$	independently characterized values reported by Hu et al.; used for a diffusion-based check of the fitted LDF coefficient
Equilibrium model	Toth reference; Langmuir baseline comparison	Toth was validated for 13X over a broad temperature range; Langmuir provides a transparent low-pressure comparison
Kinetic coefficient	k_{LDF} fitted or calculated consistently for the selected bead and gas	depends on bead radius, macropore diffusivity, temperature and carrier gas and must therefore not be treated as a universal material constant

without creating an unnecessarily large network.

4 Hybrid modelling concept

4.1 DEM packing generation

Spherical beads are inserted above a cylindrical container and settle under gravity. The final state is accepted only after the kinetic energy and mean particle velocity fall below prescribed tolerances. Independent random seeds, wall friction and particle friction generate different mechanically realizable structures. Particle number, size distribution and target bed dimensions are controlled so that stochastic topology can be studied separately from bulk porosity.

Required DEM outputs are particle centres, radii, contacts and wall contacts. Acceptance checks include particle count, size distribution, bed height, porosity, radial and axial porosity profiles, coordination number, maximum normalized overlap and reproducibility. The adsorption state does not act back on the particle motion in the first study; the geometry is frozen after packing.

4.2 Geometry-defined pore network

The final DEM geometry is converted to a radical-Delaunay tessellation, which accounts for unequal sphere radii. The dual weighted Voronoi cells define pore bodies and their shared faces define candidate throats. Each pore body α has gas volume V_α , and each throat $\alpha\beta$ has length $L_{\alpha\beta}$ and open cross-sectional area $A_{\alpha\beta}$. The initial diffusive conductance is

$$G_{\alpha\beta}^d = D_{\text{CO}_2, \text{N}_2} \frac{A_{\alpha\beta}}{L_{\alpha\beta}}, \quad (1)$$

where $D_{\text{CO}_2, \text{N}_2}$ is the binary molecular diffusivity. An optional shape factor can be introduced only after comparison with simple analytical geometries. Unlike a proximity cutoff chosen by hand, this construction derives network connectivity from the void geometry.

4.3 Where and how adsorption is implemented

Adsorption is implemented in the transient pore-network solver, not in DEM and not inside a data-driven surrogate. DEM ends after producing the fixed geometry. Each physical bead i receives a time-dependent average solid loading $q_i(t)$ in mol,kg^{-1} . The local gas concentration seen by the bead is the area-weighted concentration of its adjacent pores,

$$c_i^g = \frac{\sum_{\alpha \in \mathcal{P}_i} A_{i\alpha} c_\alpha}{\sum_{\alpha \in \mathcal{P}_i} A_{i\alpha}}, \quad (2)$$

where $\mathcal{P}_i$ contains the pore bodies adjacent to particle i and $A_{i\alpha}$ is the particle surface area associated with pore α .

For the transparent baseline, the equilibrium loading follows a Langmuir law,

$$q_i^* = \frac{q_{\max} b(T) p_i}{1 + b(T) p_i}, \quad p_i = c_i^g RT. \quad (3)$$

The primary model uses the Toth isotherm fitted to the selected 13X data,

$$q_i^* = \frac{q_m b p_i}{[1 + (b p_i)^n]^{1/n}}. \quad (4)$$

The bead-average uptake is governed by the LDF equation

$$\frac{dq_i}{dt} = k_{\text{LDF},i} (q_i^* - q_i). \quad (5)$$

For macropore-diffusion control, the fitted coefficient is checked against the spherical-particle estimate

$$k_{\text{LDF},i} \approx \frac{15 D_{e,i}}{R_i^2}, \quad (6)$$

with the precise coefficient and effective diffusivity definition kept consistent with the selected LDF derivation.

The conservative gas balance for every pore body is

$$V_\alpha \frac{dc_\alpha}{dt} = \sum_{\beta \in \mathcal{N}_\alpha} G_{\alpha\beta}^d (c_\beta - c_\alpha) - \sum_{i \in \mathcal{S}_\alpha} w_{i\alpha} m_i \frac{dq_i}{dt}, \quad (7)$$

where m_i is dry sorbent mass and $w_{i\alpha} = A_{i\alpha} / \sum_{\gamma \in \mathcal{P}_i} A_{i\gamma}$ distributes the bead uptake among its neighboring pores. The same adsorption rate added to the solid in Eq. (5)

is removed from the gas in Eq. (7). Thus, total CO_2 is conserved to time-integration tolerance.

The coupled ODE system is stiff when throat conductances and adsorption rates span wide ranges. An implicit BDF or backward-Euler method with adaptive time stepping is therefore recommended. At every step the implementation evaluates local concentrations, equilibrium loadings, LDF rates and pore sinks as one coupled residual. Sequentially updating adsorption after gas diffusion would be allowed only after a timestep-convergence study.

4.4 Reference outputs

Each network simulation provides:

- total uptake $M_{\text{ads}}(t) = \sum_i m_i q_i(t)$;
- times t_{50} and t_{90} to reach 50 and 90% of final uptake;
- axial penetration and loading profiles;
- particle-level utilization q_i/q_i^* ;
- poorly utilized fraction at a prescribed time;
- effective bed diffusivity from a separate non-adsorbing steady test;
- mass-balance error and boundary CO_2 flux.

The effective diffusivity is obtained by imposing concentrations c_h and c_l at opposite bed boundaries, solving the non-adsorbing steady network, and calculating

$$D_{\text{eff}} = \frac{\dot{N}_{\text{CO}_2} H}{A(c_h - c_l)}. \quad (8)$$

This test separates pure bed transport from adsorption kinetics.

4.5 Physical accessibility descriptors and homogeneous reduction

For a throat ij , the diffusive conductance is

$$G_{ij} = D \frac{A_{ij}}{L_{ij}}. \quad (9)$$

An effective inlet-to-interior conductance $G_{\text{in,int}}$ is calculated without adsorption by fixing the inlet-pore concentration to one and the concentration of pores at least five particle diameters into the bed to zero. Conservation is solved at all intermediate pores, and the total inlet flux for this unit concentration difference defines $G_{\text{in,int}}$. Additional descriptors measure unique second-layer pores, inlet-to-interior throat count, polar coverage, inlet pore volume and geometric tortuosity.

Initial results across the twenty accepted beds show that bulk porosity is not predictive, whereas $G_{\text{in,int}}$ has Spearman correlations of 0.968 with uptake at 5000 s and -0.959 with the underutilized-particle fraction. The unique second-layer pore count correlates with t_{50} at -0.814. Bootstrap intervals exclude zero and the corresponding univariate correlations remain significant after false-discovery-rate correction. These observations motivate predefined low-dimensional models rather than unrestricted feature selection.

The selected physical quantities will parameterize a homogeneous one-dimensional model,

$$\varepsilon_b \frac{\partial c}{\partial t} = D_{\text{eff}} \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_b \frac{\partial q}{\partial t}, \quad (10)$$

$$\frac{\partial q}{\partial t} = k_{\text{eff}}[q^*(c) - q]. \quad (11)$$

Here “solving the homogeneous bed” means solving Eqs. (10)–(11) along the bed height without individual pores or beads. Structure-dependent closure parameters are fitted to the conservative network curves and then related to the physical accessibility descriptors. The homogeneous prediction is compared against the complete network uptake curve, not only against scalar targets.

5 Proposed work packages

WP1: Material definition and controlled DEM ensemble

Select one commercial 13X dataset and define bead size, density, equilibrium and kinetic parameters consistently. Generate a small verification ensemble followed by statistically independent beds. Vary random seed, friction, polydispersity and cylinder-to-particle diameter ratio. Retain matched groups with nearly equal porosity and size distribution but different topology.

Deliverables: accepted DEM files; parameter provenance table; packing-quality report; particle and contact visualization.

WP2: Pore-network extraction and geometric verification

Construct the radical-Delaunay/Voronoi network and calculate pore volumes, throat areas, lengths and particle-pore surface associations. Verify total solid and void volume, connectivity, boundary identification and convergence with geometric tolerances. Test ordered sphere arrays and small packings with analytically understood bottlenecks.

Deliverables: reproducible DEM-to-network converter; pore/throat files; geometry verification cases.

WP3: Conservative diffusion-adsorption solver

Implement Eqs. (5) and (7). Begin with non-adsorbing diffusion, then equilibrium adsorption, and finally transient LDF kinetics. Verify mass conservation, positivity, equilibrium capacity, timestep convergence and expected scaling with bed height, molecular diffusivity and bead radius.

Deliverables: transient solver; verification report; particle and pore uptake histories; effective-diffusivity and uptake labels.

WP4: Structural mechanisms and conventional baselines

Quantify how uptake, t_{50} and poorly utilized fraction vary between matched packings. Compare porosity with inlet-to-interior conductance, second-layer pathways, polar coverage, tortuosity and throat statistics. Use bootstrap intervals, leave-one-bed-out prediction

and target-permutation tests. Determine whether the response reflects physical accessibility rather than global porosity or a boundary-labelling artifact.

Deliverables: structure–performance database; baseline models; matched-packing analysis; physical interpretation of bottlenecks.

WP5: Homogeneous model and physical closure

Solve Eqs. (10)–(11) with the same bed height, inlet concentration, isotherm, LDF coefficient and initial condition as the pore-network cases. Fit an effective transport closure separately to each reference curve. Test whether porosity alone predicts that closure and whether inlet-to-interior conductance, second-layer connectivity and polar coverage provide a reproducible improvement on withheld beds.

Deliverables: verified one-dimensional solver; fitted closure values; out-of-fold physical correlations; full-curve comparison and timing analysis.

WP6: Robustness and publication decision

Repeat the analysis for an unseen bead-size distribution and bed diameter. Quantify uncertainty across seeds and sensitivity to throat definitions, inlet labelling and the chosen five-particle-diameter accessibility depth. Repeat a limited set at another diameter ratio if resources permit. Graph learning is reserved for later work and is considered only after a substantially larger geometry ensemble exists and transparent closures have been exhausted.

Figure 2 summarizes the work-package dependencies.

6 Simulation matrix and data strategy

The completed first ensemble contains twenty independent beds with 1850 beads, $D/d_p = 10$ and identical material and operating parameters. This ensemble is sufficient for verification, uncertainty estimates and low-dimensional leave-one-bed-out baselines, but not for a high-capacity machine-learning model. An extension to another diameter ratio is used as an external structural check rather than as training augmentation. If operating conditions are varied later, all conditions belonging to one DEM realization remain in the same validation fold because they are not independent geometries.

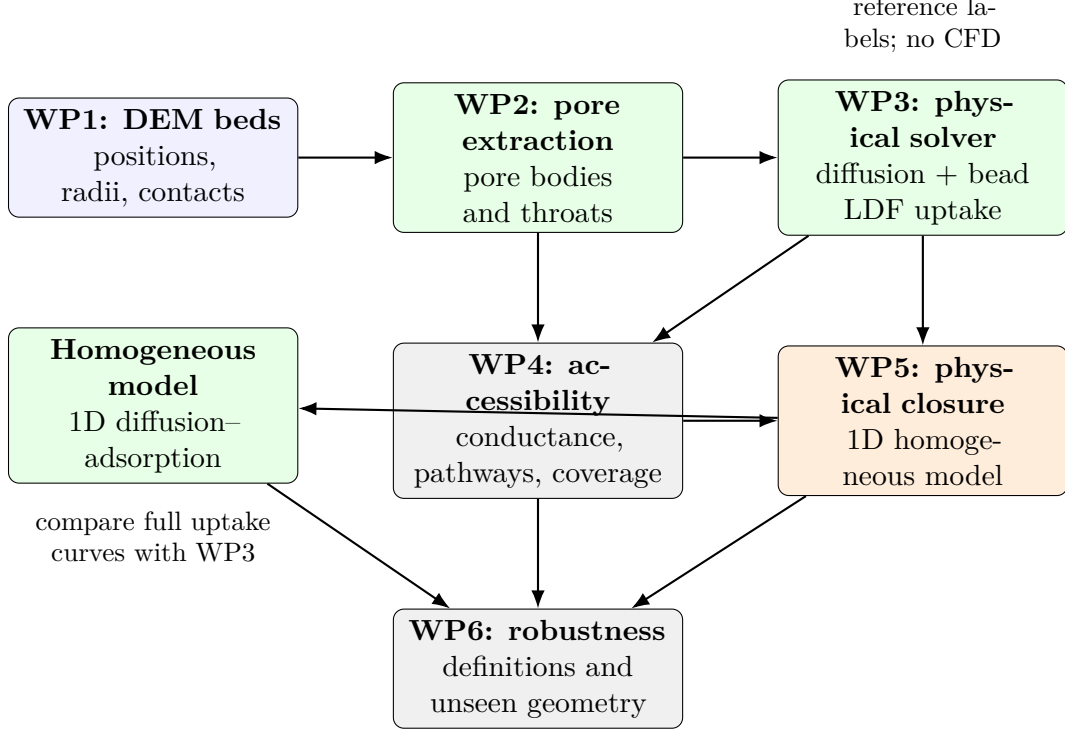


Figure 2: Block diagram of the physics-focused hybrid workflow. DEM determines a realizable geometry, the pore-network equations generate conservative reference solutions, and physical accessibility descriptors inform a reduced homogeneous model. Adsorption thermodynamics and mass conservation remain explicit.

Dimensionless groups aid transfer and interpretation:

$$Fo_b = \frac{D_{\text{CO}_2, \text{N}_2} t}{H^2}, \quad Da_b = \frac{k_{\text{LDF}} H^2}{D_{\text{CO}_2, \text{N}_2}}, \quad \Pi_q = \frac{(1 - \varepsilon_b) \rho_b q^*}{\varepsilon_b c_{\text{in}}}. \quad (12)$$

These distinguish bed-diffusion time, bead uptake time and adsorption capacity relative to gas inventory.

7 Validation, baselines and publication logic

7.1 Mandatory verification

The numerical solver must satisfy:

1. conservation of gas plus adsorbed CO_2 against integrated boundary flux;
2. nonnegative concentrations and $0 \leq q_i \leq q_i^*$ for an adsorption step;
3. convergence under timestep reduction;

4. reproduction of equilibrium capacity in a closed uniform system;
5. linear scaling of non-adsorbing flux with concentration difference;
6. correct H^2/D diffusion-time scaling;
7. invariance to graph node ordering.

Numerical verification does not establish physical validity. Physical checks include comparison of the single-bead LDF response with published 13X uptake data, comparison of effective diffusivity with accepted porous-medium bounds, and sensitivity to throat-area definitions. If experiments become available, the strongest validation is transient mass uptake of a small packed column under quiescent boundary exposure.

7.2 Mandatory reduced-model baselines

The physical closure is compared with:

- porosity only;
- raw inlet-pore count only;
- inlet-to-interior conductance only;
- unique second-layer pore count only;
- conductance combined separately with tortuosity and polar coverage;
- a predefined three-descriptor physical model; and
- direct pore-network solution as the conservative reference.

Primary metrics are out-of-fold R^2 and normalized RMSE for scalar targets, integrated absolute error of uptake curves, error in t_{50} and t_{90} , mass-balance error of the deployed hybrid model and CPU time. Confidence intervals are calculated across complete packings. With only twenty beds, the number of fitted descriptors is deliberately kept small.

7.3 Success and failure criteria

The study supports a physical-closure claim only if:

1. matched beds show statistically meaningful variability beyond porosity and bead-size variation;
2. predefined accessibility descriptors improve unseen-packing predictions over porosity and raw inlet count;
3. the fitted closures reproduce full homogeneous-model uptake curves, not only one scalar target; and
4. the advantage persists for an unseen geometry or larger bed.

If the first condition fails, packing topology is not important in the selected regime. If the second fails, the simplest inlet descriptor is preferable. If the homogeneous model cannot reproduce the reference curves even with fitted closures, the reduced model is structurally inadequate and must be extended.

8 Expected contributions and limitations

The expected contributions are:

1. a reproducible conversion from DEM-generated 13X beds to a geometry-defined pore graph;
2. a conservative particle-specific diffusion-adsorption network with explicit LDF uptake;
3. quantitative evidence of whether equal-porosity beds differ in sorbent accessibility;
4. an interpretable accessibility-based closure for a cheap homogeneous model; and
5. a quantitative distinction between entrance distribution and deeper-bed transport limitations.

The first study neglects forced flow, pressure drop, heat of adsorption, competitive water adsorption, particle deformation during operation and adsorption-induced changes

in material properties. Zeolite 13X is strongly sensitive to water, so conclusions apply initially to dry CO_2/N_2 conditions. Heat effects can also become important at higher CO_2 loading; the isothermal assumption should therefore be limited to dilute conditions or small beds with effective thermal control. These limitations are stated rather than hidden because the proposed contribution is structural reduction, not a complete industrial PSA model.

9 Conclusion

The proposed paper combines DEM, a pore-scale resistance network and a physical homogeneous reduction without using CFD. DEM supplies realistic zeolite-bead packings; the geometry-derived pore network solves gas diffusion; and one explicitly coupled LDF equation per particle describes CO_2 adsorption. Physical accessibility descriptors are used to construct a low-dimensional closure for a homogeneous adsorption model. Commercial zeolite 13X beads provide a well-supported first material because their size, selectivity and macropore-controlled kinetics are documented and compatible with spherical DEM. The decisive scientific test is whether inlet-to-interior accessibility explains differences among bulk-matched beds that porosity misses and whether that information transfers to a homogeneous model. This framing preserves physical meaning, avoids the cost of CFD and provides a clear path to later pressure-driven, non-isothermal or graph-learning extensions.

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